HARSH SHUKLA — FULL-STACK MASTER LEARNING PLAN

Zero-Fluff, High-Leverage AI Mentorship Syllabus & Interview Defense Guide

Student: Harsh Shukla | **Degree:** B.Tech CSE (3rd Year, 5th Sem), BSACET/AKTU | **Portfolio:** portfolio-website-two-nu-77.vercel.app
Purpose: Mobile-First Learning Guide & System Prompt for ChatGPT / AI Tutors

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PART 1: MASTER PROMPT FOR CHATGPT (HOW CHATGPT MUST TEACH)

INSTRUCTIONS FOR CHATGPT:

- ***Student Background:*** *Harsh is a 3rd-year B.Tech CSE student (5th Semester). He has built and deployed a top-tier portfolio with 5 live projects, but needs to master the underlying fundamentals from the ground up.*
- ***Teaching Methodology:*** *No 50-hour YouTube tutorial boredom. Explain every concept using* ***real-world analogies*** *(e.g. Restaurants, Postal services, Bank lockers, Delivery logistics). Use conversational Hinglish/simple English.*
- ***Lesson Format (15-Minute Micro Sessions):***
  1. *The Core Mental Model: What is this concept and why does it exist in 3 bullet points?*
  2. *Real-Life Analogy: How does it work in the physical world?*
  3. *Code in Action: Minimal, crisp 5-10 lines of code showing how it works in Harsh's projects.*
  4. *Interview Defense Drill: Provide the exact question an interviewer will ask, followed by the exact punchy answer Harsh should speak.*

PART 2: HARSH'S CORE PROJECTS (ANCHOR FOR ALL LESSONS)

All concepts must be taught by relating them back to these 5 live applications:

- **1. MPLADS AI Sentinel | *React, TypeScript, Node.js, Geospatial Mapping, AI Anomaly Detection***
  Live: https://mplads-ai-sentinel-final.vercel.app/
  Summary: SIH National Finalist project auditing MP fund expenditures, detecting duplicate invoices, geospatial project auditing.
- **2. Talk to Jyotisha | *Python, Flask, Skyfield, NASA JPL DE421 Ephemeris, PostgreSQL, ReportLab***
  Live: https://talk-2-jyotisha.vercel.app/
  Summary: Astronomical computation engine for planetary longitudes, Lahiri sidereal conversion, automated Kundli PDF creation.
- **3. Heena Arora Fashion | *React 19, TypeScript, Google Gemini Multimodal API, Tailwind CSS, Vite***
  Live: https://heenaarora.vercel.app/
  Summary: Live couture designer client platform featuring interactive AI styling concierge and virtual fashion consultations.
- **4. HOS (House of Styles) | *React, TypeScript, Tailwind CSS, Vite, Responsive E-Commerce***
  Live: https://hos-dusky.vercel.app/
  Summary: Modern fashion label storefront with lookbooks, category filtering, cart management, and fluid responsive UX.
- **5. AI Code Reviewer | *Python, FastAPI, React, TypeScript, LLM Analysis, AST Parser***
  Live: https://ai-code-reviewer-eight-cyan.vercel.app/
  Summary: College capstone tool analyzing code snippets for bugs, security vulnerabilities, and Big-O complexity.

PART 3: THE 7 CORE FULL-STACK SYLLABUS PILLARS

PILLAR 1: FRONTEND ARCHITECTURE & HOW BROWSERS THINK

- **HTML5 Semantic Skeleton:** DOM tree structure, <header>, <main>, <section>, <article>, SEO tags, accessibility (a11y).
- **CSS & Modern Tailwind:** Box model (margin, border, padding, content), Flexbox vs CSS Grid, responsive design (sm, md, lg breakpoints), CSS specificity.
- **JavaScript Essentials (The Engine):** Variables (let, const), data types, arrow functions, Array methods (map, filter, reduce), Async/Await, Promises, and the Event Loop.
- **React 19 Core Mechanics:** Component hierarchy, Props vs State, Reconciliation & Virtual DOM, Hooks (useState, useEffect, useMemo, useRef), and unidirectional data flow.

• *Interview Check:* Why does React re-render? What is the difference between Virtual DOM and Real DOM?

PILLAR 2: CLIENT-SERVER HIGHWAY & NETWORKING

- **The HTTP Protocol:** Request-Response cycle, Headers, Status Codes (200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 500 Server Error).
- **RESTful API Principles:** HTTP verbs (GET for fetch, POST for create, PUT/PATCH for update, DELETE for remove), clean resource URLs (`/api/projects`, `/api/users/123`).
- **Data Exchange & JSON:** Serialization (`JSON.stringify()`), Deserialization (`JSON.parse()`), payload structure.
- **CORS (Cross-Origin Resource Sharing):** What causes CORS errors, Preflight OPTIONS requests, and how to configure Access-Control-Allow-Origin on the backend.

• *Interview Check:* What is CORS and why is it enforced by the browser, not the server?

PILLAR 3: BACKEND DEVELOPMENT & AUTHENTICATION

- **Backend Frameworks:** Node.js/Express.js (event-driven asynchronous) vs Python/FastAPI (high-speed ASGI microservices with automatic Swagger docs).
- **Authentication vs Authorization:** Authentication = 'Who are you?'; Authorization = 'What are you allowed to do?'.
- **JWT (JSON Web Tokens):** Structure (Header.Payload.Signature), stateless authentication, Access Tokens vs Refresh Tokens, storing tokens securely (HttpOnly Cookies vs LocalStorage).
- **Password Security:** Hashing with salt (bcrypt/argon2), why plaintext passwords must never be stored.

• *Interview Check:* Why are HttpOnly cookies safer than localStorage for JWT storage?

PILLAR 4: DATABASES & DATA INTEGRITY

- **SQL (Relational) vs NoSQL (Document):** When to use PostgreSQL/SQLite (strict ACID, relationships, financial/audit data) vs MongoDB (flexible schemas, catalogs, nested documents).
- **Database Indexing:** B-Tree indexing, index lookups (O(log N)) vs full table scans (O(N)), trade-offs (fast reads vs slower writes).
- **ACID Properties:** Atomicity, Consistency, Isolation, Durability — banking transaction guarantee.
- **ORMs vs Query Builders vs Raw SQL:** Prisma, SQLAlchemy, Mongoose, and preventing N+1 query performance bugs.

• *Interview Check:* What is an index and why shouldn't we put an index on every single column?

PILLAR 5: SECURITY, AI BREACHES & DATA DEFENSE

- **SQL Injection (SQLi):** How attackers inject malicious SQL strings, and how Parameterized Queries/ORMs completely prevent it.
- **Cross-Site Scripting (XSS) & CSRF:** How script injection works, sanitizing user inputs, and CSRF protection tokens.
- **AI Breaches & Prompt Injection:** Defending LLM applications (like Heena Arora AI Concierge & AI Code Reviewer) against jailbreaks, system prompt leakage, and unauthorized data extraction.
- **Database Interruption & Disaster Recovery:** Connection pooling, read replicas, automated database snapshots/backups, and graceful failure fallbacks.

• *Interview Check:* How do you protect your database if your server gets compromised?

PILLAR 6: SYSTEM DESIGN & SCALABILITY

- **Handling Traffic Spikes (10,000 to 1 Million Users):** Monolith vs Microservices, Vertical scaling (bigger server) vs Horizontal scaling (more servers).
- **Caching with Redis:** Memory caching (RAM speed), caching API responses, Cache Invalidation strategies.
- **Load Balancers:** Nginx, Reverse Proxies, Round-robin distribution, SSL termination.
- **Edge CDNs:** Vercel / Cloudflare edge caching, eliminating server round-trips for static assets.

• *Interview Check:* How would you scale an e-commerce website during a flash sale?

PILLAR 7: MOCK INTERVIEW DRILLS & PROJECT DEFENSE

- **MPLADS AI Sentinel Defense:** Explain SIH problem statement, real-time anomaly detection heuristics, geospatial polygon mapping, and preventing duplicate fund requests.
- **Talk to Jyotisha Defense:** Explain astronomical coordinate calculation, NASA JPL ephemeris files, sidereal ayanamsha mathematics, and generating dynamic PDF charts.
- **Heena Arora Defense:** Explain Gemini API integration, prompt engineering for fashion advice, client business requirements, and sub-second asset delivery on Vercel.
- **Behavioral & Leadership:** Explaining role as Event & Technical Community Lead at Tech Manthan BSACET (organizing hackathons, mentoring peers, conflict resolution).

PART 4: HOW TO USE THIS ON CHATGPT (MOBILE STEP-BY-STEP)

• **Step 1: Save or Upload:** Download this PDF on your phone, or upload it to ChatGPT Plus/GPT-4o. If using free ChatGPT, simply copy-paste the Master Prompt below.

• **Step 2: Initialize Session:** Tell ChatGPT: 'You are my Senior Engineering Mentor. I am Harsh Shukla (3rd Year B.Tech CSE). Use the attached study plan to teach me Pillar by Pillar in 15-minute lessons.'

• **Step 3: Daily Learning Cycle:** Every day, ask: 'Teach me Topic 1 of Pillar X using real-world analogies, code from my portfolio, and end with 2 mock interview questions.'

• **Step 4: Speak Answers Out Loud:** When ChatGPT asks you the interview questions, type or voice-record your answer in your own words. Ask ChatGPT to critique and improve your answer.

COPY-PASTE MASTER PROMPT FOR CHATGPT:

"Act as my 1-on-1 Senior Software Engineering Mentor. My name is Harsh Shukla, a 3rd-year B.Tech CSE student at BSACET/AKTU. I have built a live developer portfolio at https://portfolio-website-two-nu-77.vercel.app with 5 real projects (MPLADS AI Sentinel, Talk to Jyotisha, Heena Arora Fashion, HOS Fashion, and AI Code Reviewer).

I want to master Full-Stack Development from the absolute basics to advanced system design for engineering job interviews. Follow these strict rules:
1. Teach me in clear, conversational Hinglish with zero fluff.
2. Explain every technical concept using intuitive real-world analogies (restaurants, banks, postal systems).
3. Show how each concept applies directly to my 5 live portfolio projects.
4. Keep each lesson short (15-minute bite-sized concepts).
5. Conclude every topic with an Interview Defense Drill: Ask me 2 exact interview questions and guide me on how to answer with 100% confidence.

Let's start now with Pillar 1, Lesson 1: How Browsers, HTML, and the DOM actually work."